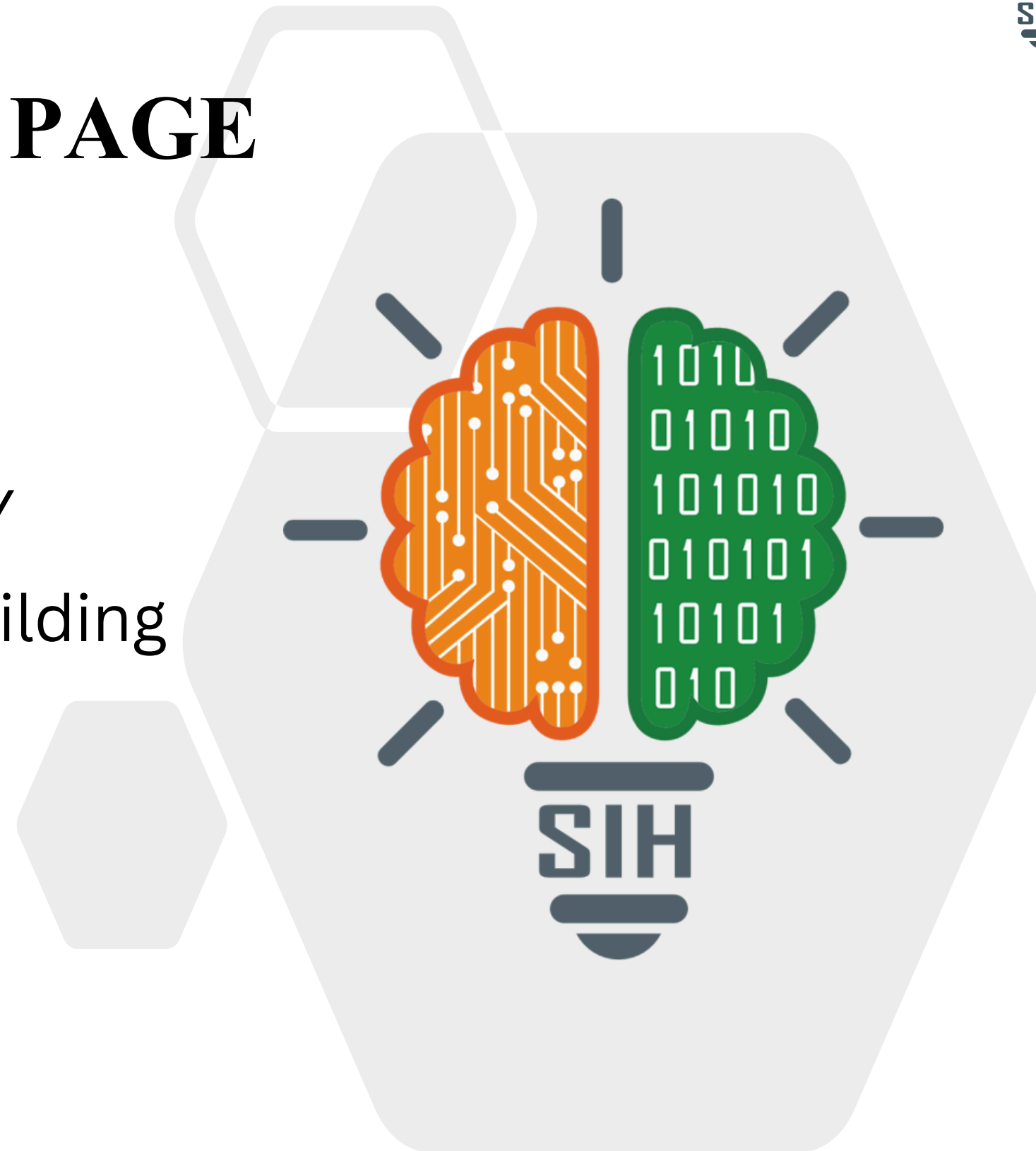


TITLE PAGE

- **Problem Statement ID – 26075**
- **Problem Statement Title-** CAPACITY
CONNECT – A Digital Capacity Building
and Learning Management Portal
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Sankalp**



CAPACITY CONNECT

Problem

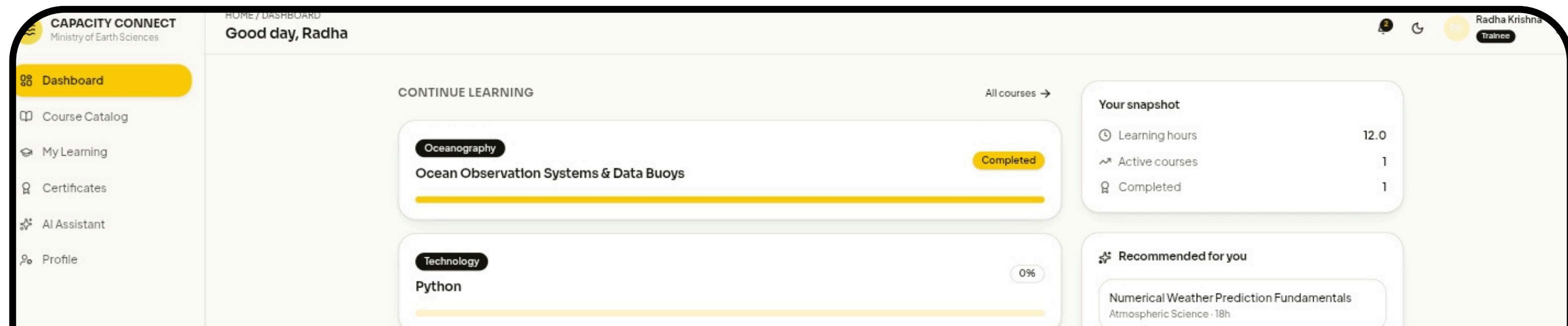
- Fragmented training records
- Manual competency tracking
- Lack of personalized learning
- No analytics
- Poor knowledge sharing

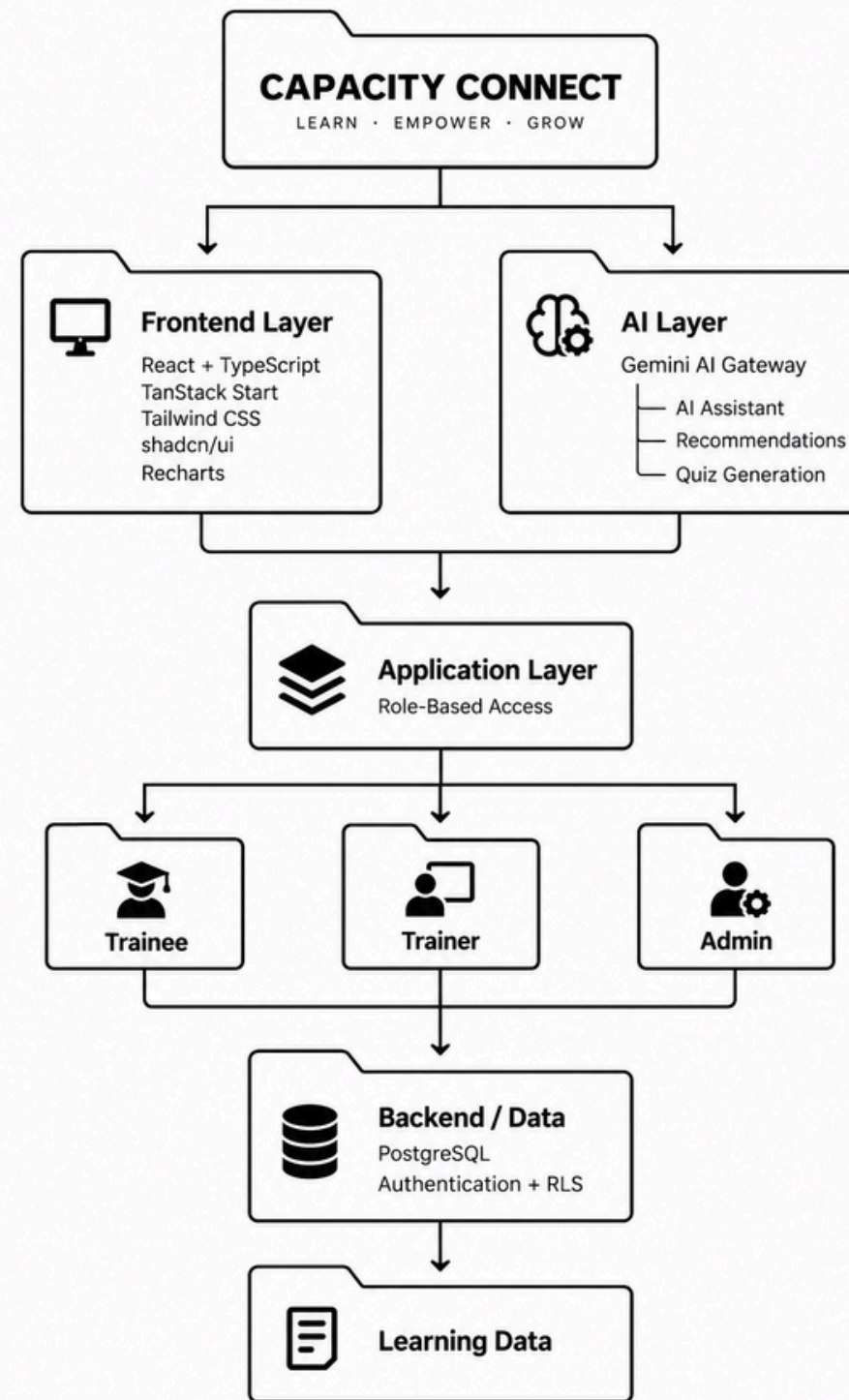
Our Solution

- Unified AI-powered LMS
- Competency Mapping
- Personalized Learning Paths
- Knowledge Repository
- AI Chatbot
- Smart Analytics

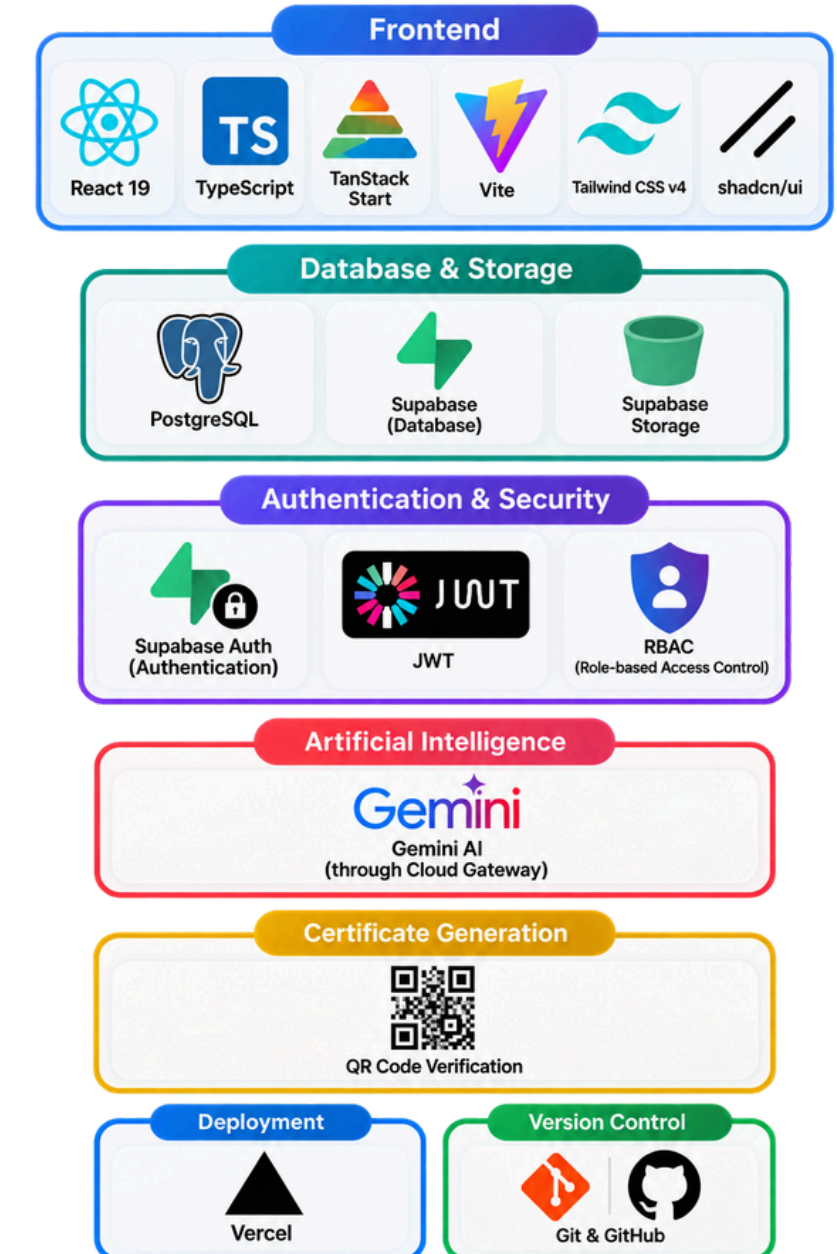
Innovation

- AI Skill Gap Analysis
- Semantic Search
- Knowledge Graph
- AI Recommendations
- Predictive Analytics





- *Frontend: Built with React 19, TypeScript, TanStack Start, Vite, Tailwind CSS v4, and shadcn/ui.*
- *Database & Storage: Supabase (PostgreSQL + Supabase Storage) for scalable database management, secure file storage, real-time data synchronization, and built-in authentication support.*
- *Authentication & Security: Supabase Auth, JWT, RBAC*
- *Artificial Intelligence: Integrated with Gemini through the cloud Gateway.*
- *Certificate Generation: QR Code Verification*
- *Deployment : Vercel*
- *Version Control: Git & GitHub*



Technologies

- Modern web architecture
- Cloud-hosted PostgreSQL
- Modular React architecture
- API-based AI integration
- Responsive web application



Security

- JWT-based sessions
- Row-Level Security
- Dedicated role table
- `has_role()` security function
- Audit logging
- Protected routes

Operational

- Three clearly defined user roles
- Trainers can manage courses
- Administrators can manage users and institutional data
- Learners have self-service learning

Challenges → Mitigation



Data privacy

Protecting sensitive student, faculty, and institutional data.



Authentication + RLS + role-based authorization

Secure access with authentication, row-level security (RLS), and role-based permissions.



AI reliability

Ensuring accurate, safe, and unbiased AI responses and content.



Structured AI workflows + human-controlled learning content

Use well-defined AI workflows and human oversight for learning materials.



Different user needs

Students, teachers, administrators, and staff have diverse requirements.



Role-specific dashboards and permissions

Tailored dashboards and access controls for each user role.



Assessment scalability

Creating and managing assessments for large numbers of users.



AI-assisted quiz generation

Automatically generate high-quality quizzes to scale assessments.



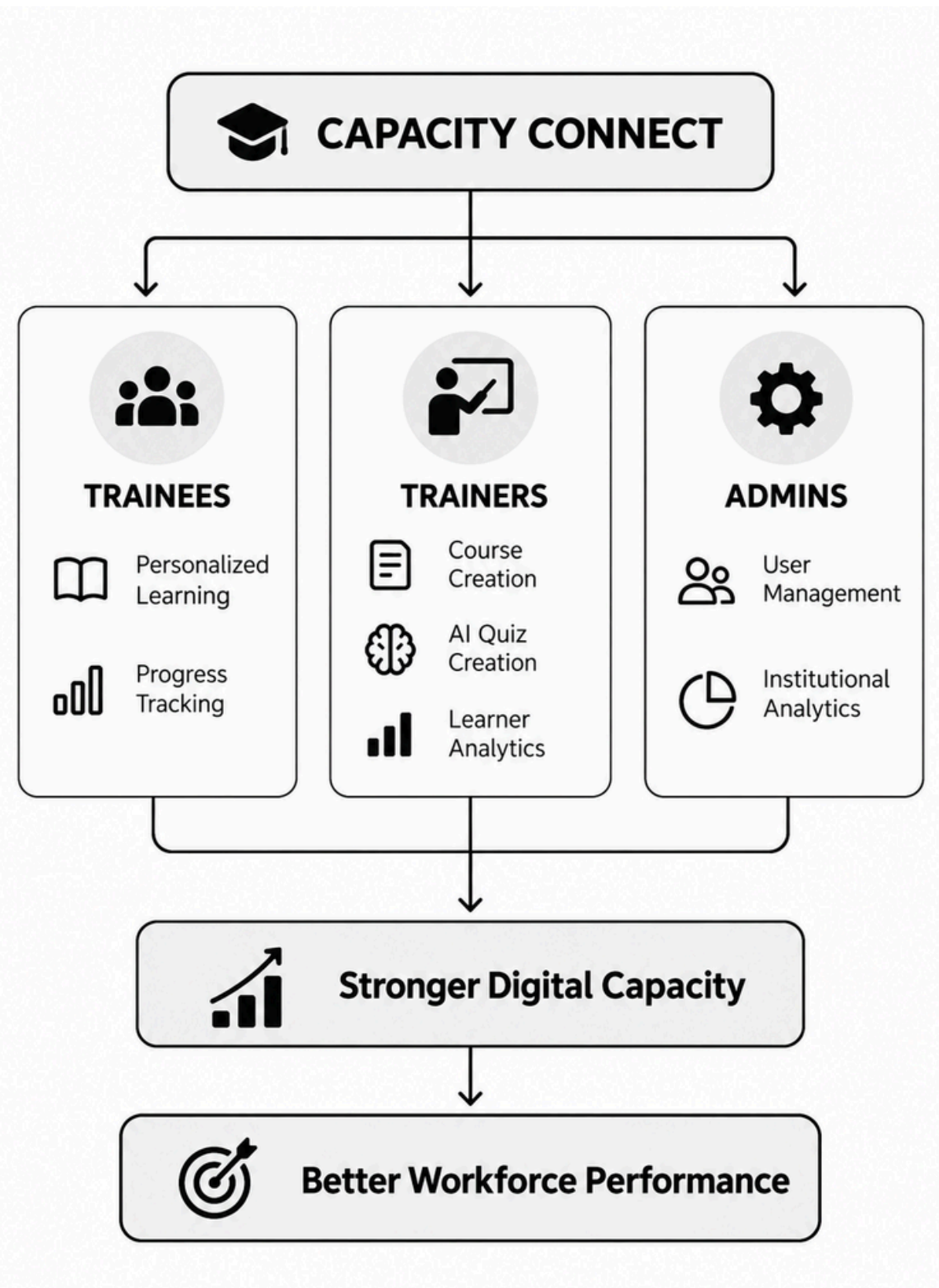
Institution-wide monitoring

Tracking usage, performance, and learning outcomes across the institution.



Centralized analytics

Unified analytics dashboard for real-time, institution-wide insights.



For Trainees:

- Personalized learning
- Self-paced courses
- Timed assessments
- Progress tracking
- Certificates
- AI assistant

For Trainers:

- Course creation and management
- AI-generated quizzes
- Learner progress monitoring
- Course analytics

For Admin:

- User management
- Role assignment
- Institutional analytics
- Centralized oversight

**One platform → Every learner →
Personalized growth →
Measurable institutional
capacity**

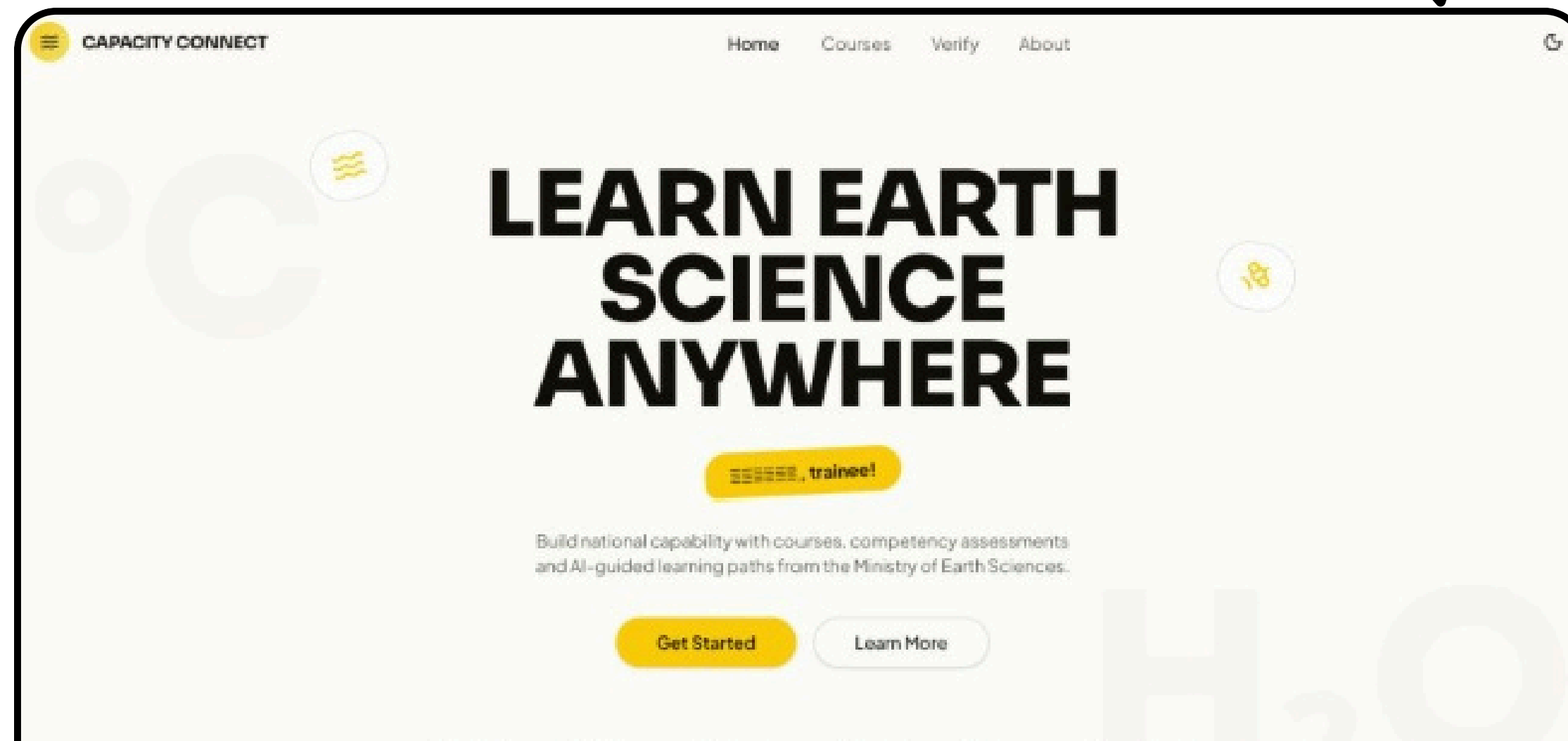
GitHub repository: github.com/radhakrishna131/Hackathon

Prototype Live Demo: <https://sankalp-sih.vercel.app>

Reference Taken: <https://www.techsnap.in>

Research:

- Perplexity
- Google
- Gemini
- Reddit
- youtube



For Features reference

- Swayam
- Scaler
- npTEL
- FreeCodeCamp
- coursera